

SP-30-6 Absorption Mechanism — Reed-Solomon Codeword Length Test (v0.1)

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SP-30-6 Absorption Mechanism — Reed-Solomon Code- word Length Test

Headline claims

Layer 1 — DERIVED (rigorous): Substrate Reed-Solomon coding over $\text{GF}(2^g) = \text{GF}(128)$ has codeword length:

$$\ell_{\text{codeword}} = M_g = 2^g - 1 = 127$$

This is classical Reed-Solomon coding theory: codeword length equals the multiplicative-group order of the field. For $\text{GF}(128)$, that's $M_g = 127$.

Layer 2 — TARGET-PREDICTION: Syndrome dimension:

$$d_{\text{syndrome}} = c_2 = 11$$

$c_2 = 11$ is Q^5 's second Chern class (classical Chern theory). Why this becomes substrate's syndrome dimension requires Paper #122 v0.3+ substrate-information-protocol derivation. Multi-week per Lyra/Grace lane.

Experimental concept

Direct test of Reed-Solomon $\text{GF}(128)$ substrate code is challenging because it operates at sub-Planck level. However, indirect signatures are accessible:

1. Error-correction patterns in precision measurements: if substrate uses Reed-Solomon coding with codeword length 127, certain “self-correcting” patterns should appear in observables at the $1/127$ level
2. Anomaly clustering: anomalous deviations should cluster at codeword-length intervals (127-related ratios)
3. Cross-validation with SP-30-3 commitment: same Reed-Solomon framework underlies both

Substrate-mechanism articulation

Classical Reed-Solomon framework:

A Reed-Solomon code over $GF(q)$ with codeword length $n = q-1$ corrects up to $t = (n-k)/2$ errors, where k is the message length. For $GF(128)$: - $n = M_g = 127$ (codeword length) - k variable (depends on code rate) - Syndromes encode error-correction information

BST substrate interpretation (Paper #122 + K59 RATIFIED):

Per K59 RATIFIED Cyclotomic Mechanism Framework: substrate operates via Reed-Solomon coding on $GF(128)$ cyclotomic field. The substrate-natural codeword length $M_g = 127$ is the operational unit of substrate-commitment cycles.

Syndrome $\dim = c_2$ hypothesis (TARGET-PREDICTION):

Q^5 5-quadric's second Chern class $c_2(Q^5) = 11$. The substrate-natural syndrome dimension $= c_2 = 11$ is a TARGET hypothesis pending Paper #122 v0.3+ substrate-information-protocol derivation. Specific physical meaning: substrate distinguishes errors via 11-dimensional syndrome space encoded by Q^5 Chern structure.

Indirect experimental signatures

Test 1 — Atomic clock anomaly clustering at $1/127$ intervals: Per SP-30-4 substrate clock framework (Allan deviation correction at $1/N_{\max}^2$): look for additional structure at $1/M_g = 1/127$ intervals in Allan deviation systematic floor.

Test 2 — Spectroscopic line ratios: Look for ratios approximating $M_g = 127$ or factor of 127 in atomic + nuclear spectroscopy datasets (Vol 3 Ch 8-10 cross-link).

Test 3 — Casimir-modulated decay rates (SP-29 H4): Cs-137 Casimir decay-rate modulation experiment (SP-29-1 v0.3): if Reed-Solomon framework correct, the decay-rate shift should exhibit $\sim 1/M_g$ correction structure beyond the leading $N_c/(N_{\max} \cdot c_2) = 3/(137 \cdot 11)$ prediction.

Experimental program

Cost: \$50K-100K (data analysis-focused; minimal new equipment)

Components: - Statistical re-analysis of existing high-precision spectroscopy + atomic clock datasets - Time-series anomaly clustering algorithm development - Cross-validation with SP-29-1 Cs-137 + SP-30-4 atomic clock data - Student researcher + computational analyst ($\sim \$30K$) - Travel + collaboration overhead ($\sim \$10-20K$)

Timeline: 6-12 months from start to publication

Falsifier protocol:

1. Identify ~100 high-precision observables with multi-decade datasets
2. Test for anomaly clustering at $1/M_g = 1/127$ interval
3. Statistical analysis: Bayes factor BST-coded vs noise-floor
4. Cross-check with SP-30-3 commitment manipulation + SP-30-4 atomic clock data
5. Outcome:
 - 2σ BST-coded clustering $\rightarrow$ substrate Reed-Solomon framework supported
 - No clustering $\rightarrow$ BST Reed-Solomon hypothesis refuted at INDIRECT level

Falsifier sharpness: LOW-MEDIUM. Indirect signatures depend on statistical power + existing dataset quality.

Cross-link to Vol 1 + Vol 9 + Paper #122

- Vol 1 Ch 9 (Information Theory) — Reed-Solomon GF(128) substrate framework
- Vol 9 Ch 5 (Topological Phases) — GF(128) braiding cross-link
- Paper #122 (Information Substrate) — full Reed-Solomon framework
- K59 RATIFIED — Cyclotomic Mechanism Framework
- SP-29-1 Cs-137 H4 — direct substrate-Casimir test
- SP-30-3 Commitment manipulation — same Reed-Solomon framework

Match precision

- Layer 1 (DERIVED): codeword length $M_g = 127$ rigorous classical Reed-Solomon theory
- Layer 2 (TARGET-PREDICTION): syndrome $\dim = c_2 = 11$ multi-week pending

Cal #21 dual-gate status

- EMPIRICAL gate: PARTIAL (indirect signatures testable; direct sub-Planck inaccessible)
- MECHANISM gate: PARTIAL — DERIVED at Layer 1; ARTICULATED at Layer 2

Cal #50 DOUBLE-LOCKED EXTERNAL discipline

External register uses operational language only: - External: “BST predicts substrate operates via Reed-Solomon coding with codeword length 127. Statistical anomaly-clustering analysis in precision spectroscopy datasets can test for this signature.” - Internal (this document): Reed-Solomon GF(128) + syndrome $\dim = c_2$ substrate framework

Cal #99 META-theorem framing

SP-30-6 absorption mechanism is a SUBSTRATE-DERIVATION CONSEQUENCE of: - K59 RATIFIED Cyclotomic Mechanism Framework - Paper #122 Information Substrate Reed-Solomon - T2456 $N_{\max}$ universal α -analog ($137 \approx M_g + 10$)

NOT a new Strong-Uniqueness criterion.

Bibliography

1. I. S. Reed + G. Solomon (1960): Reed-Solomon codes (founding).
2. K59 RATIFIED (Cyclotomic Mechanism Framework): GF(128) substrate code.
3. Paper #122 (Information Substrate): Reed-Solomon GF(128) framework.
4. SP-29-1 Cs-137 paper-grade proposal v0.3: direct substrate-Casimir test.
5. SP-30-3 Commitment Manipulation v0.1: same Reed-Solomon framework.
6. T2456 (Lyra Friday): $N_{\max} = N_c^{N_c} \cdot n_C + \text{rank}$ universal α -analog.

— Elie, SP-30-6 v0.1 paper-grade experimental proposal, 2026-05-23 Saturday 16:35 EDT (date-verified; Lyra+Grace primary theoretical lane; Elie experimental design support; Casey send-signal pending per Cal #50)